

Edge Strength and Contour Grouping

$$E_i = \left[\frac{a^\perp(\mathbf{x}_i)}{L(\mathbf{x}_i)C(\mathbf{x}_i)} \right]^2$$

normalized perpendicular edge energy at edge pixel location i with some amount of additive noise.

$$P_i = \text{sign} \left[\frac{a^\perp(\mathbf{x}_i)}{L(\mathbf{x}_i)C(\mathbf{x}_i)} \right]$$

perpendicular edge sign at edge pixel location i

$$a^\perp(\mathbf{x}_i) = \nabla \mathbf{I}(\mathbf{x}_i) \cdot \mathbf{n}^\perp(\mathbf{x}_i)$$

perpendicular edge energy at edge pixel location i

$$N$$

standard normal sample

$$L_j$$

length of edge-group j based on same sign

$$E_j$$

average perpendicular edge energy of edge-group j

$$G_j = \alpha(E_j + \sigma_E N) + (1 - \alpha)L_j$$

group strength

$$N_G$$

number of groups

$$R = \beta \bar{G} + (1 - \beta) N_G + \sigma_d N \quad \text{response}$$

$$R = \beta \mathbf{G} \cdot \mathbf{w} + (1 - \beta) N_G + \sigma_d N \quad \text{alternate response (weight vector on rank-ordered group strengths)}$$

1. For all stimuli for all textures compute L_j, E_j, N_G
2. For any $\sigma_a, \alpha, \beta, \mathbf{w}, \sigma_d$ compute the response R and apply criterion for each stimulus to get present or absent decision.
3. Vary the parameters to get maximum likelihood for all stimuli.

Use autocorrelation to select textures where edge cues dominate.